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AGENTIC FLASHLIGHT CO

Agents handle routine operations
Humans govern strategy, exceptions, and high-stakes decisions

Build vs Buy Matrix

BUY (SaaS Tools)

Stripe/ Revolut	ShipStn	Zendesk
DataDog / PowerBI	DataDog / PowerBI	Xero

Commodity | PCI Handled | Fast

BUILD (Custom Moat)

Orchestration (RLM)	Context Graph	Transaction Firewall
Custom Agents	Pricing Engine	Security Observer

IP Moat | Decision Trace| Control

SaaS: **Tools behind a gateway (not autonomous agents)**

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Why Agentic + What Drove the Physical Architecture

WHY AN AGENTIC FRAMEWORK?

Multi-agent, centrally governed
• Specialists coordinated centrally
• Keeps context without losing control

Commerce requires action, not answers
• Availability → checkout → pay → refund
• Transaction firewall + idempotency

Auditability by design
• Bi-temporal logs + decision provenance
• Security Observer is read-only evidence

CRITERIA THAT SHAPED THE PHYSICAL MODEL

Sovereignty + data residency
• PII + sensitive IP stay in colo
• Burst compute to cloud for elasticity/cost

Security + governance as built-in primitives
• Zero-trust posture (assume prompt injection)
• Policy-as-code (ABAC), step-up auth, HIL>$250

Compliance-ready evidence
• ISO 27001 / ISO 42001 / EU AI Act / PCI DSS
• Append-only decision trail for audits

Bottom line: agentLumen is a safe, auditable commerce agent architecture that scales via cloud without sacrificing sovereignty, IP protection, or payment-grade controls.

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Logical ⇒ Physical Mapping

Business Driver	Architecture Decision	Physical Placement
Protect IP + secrets	Custom agents (not SaaS)	COLO
Audit trail for compliance	Bi-temporal context graph	COLO (PII zone)
Prevent AI misuse	Transaction Firewall	COLO (isolated)
Elastic customer traffic	Managed storefront + CDN	CLOUD
Reduce PCI scope	SaaS payments (Stripe)	EXTERNAL
Cost efficiency	Hybrid 70/30 split	COLO + CLOUD

BUILD (IP Moat)

Orchestrator (RLM)
Domain Agents
Transaction Firewall
Security Observer

DEPLOY (Managed)

Neo4j, Qdrant
PostgreSQL, Redis
API Gateway

BUILD (IP Moat)

Stripe, ShipStation
Zendesk, Xero, Intercom
DataDog, Cloudflare

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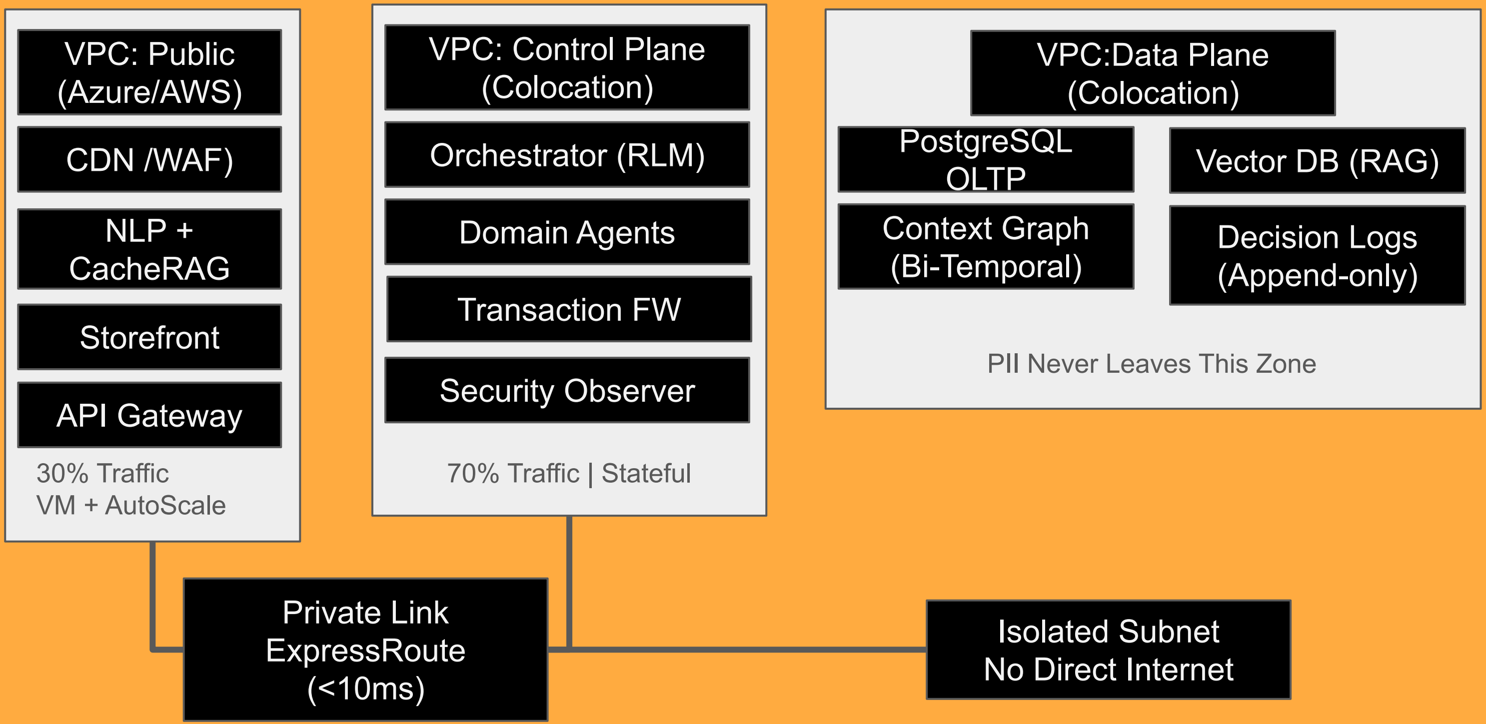
Business Architecture → Data/App/Tech Architecture → Implementation

Hybrid Deployment + Network Segmentation



Data Placement: Where everything lives

COLOCATION
Data Plane

PostgreSQL (OLTP)
- Orders, Customers
- Inventory, Products

Context Graph (Neo4j)
- Bi-temporal trace
- Decision provenance

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- Bi-temporal trace
- Decision provenance

PII + Sensitive

COLOCATION
Control Plane

Redis (Session 3h)
QDrant / Pinecone (Vec)
Neo4j (Context Graph)

Agent Runtime (GPU)
Orchestrator
Transactional FW

Security Observer

Compute + Memory

CLOUD (AZURE / AWS)
Analytics

BigQuery (OLAP)
- Aggregated only
- No PII
- RAGAS eval metrics

DataDog (SIEM/Monitoring)
- Logs Alerts
- Redacted Traces

PowerBI (Dashboards)
- Business KPIs

Analytics + Monitoring

RULE: **PII never leaves colo · Analytics gets aggregates only · Logs are redacted**

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SECURITY + COMPLIANCE ARCHITECTURE

ZERO-TRUST AGENT MODEL

"Every agent assumed compromised"

- Microsegmented VPCs
- No agent-to-agent direct calls
- All comms via Orchestrator
- All comms via Orchestrator

WRITE SAFETY (TRANSACTION FW)

AGENTS Propose → Firewall Approve → SaaS Execute

- ABAC policy enforcement
- > $250 → Human approval
- Idempotency keys
- Step-up authentication

SECURITY OBSERVER (Read-Only)

ZERO write scopes

- Watches all tool calls
- MITRE ATLAS threat tagging
- Prompt injection detection
- Supply-chain anomaly alerts

Observer Recommends | Policy Router enforces

COMPLIANCE EVIDENCE

Bi-temporal Decision Trace

- What AI knew at decision time
- What evidence was retrieved
- What action was taken
- Policy version applied

Supports: ISO 42001, NIST AI RMF, EU AI Act,

RESILIENCE + IMPLEMENTATION

WHEN AI FAILS (Graceful Degradation)

"Every agent assumed compromised"

Agent Error → Auto-retry (3x) → Rule-based fallback → Assist mode → Human escalation

Low RAG confidence	→	Corrective RAG (broaden + verify)
Prompt injection	→	Block + alert Security Observer
High-risk decision	→	Route to Transaction Firewall HITL

IMPLEMENTATION PHASES

PHASE 1 (4wk)
Infra + Auth
3 VPCs + Link
Firewall MVP

PHASE 2 (4wk)
Integrations
RAG + Webhooks
RAGAS > 0.8

PHASE 3 (4wk)
Launch + Scale
Beta → Prod
< 5% human

TOGAF | **Architecture Roadmap aligned to Business Capability increments**

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NEXT STEPS